Multidimensional Analysis of Social Discourse on X around Venezuelan Migration in Colombia

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Abstract

Colombia has become the principal host of Venezuelan migrants in Latin America, prompting a complex mix of solidarity, anxiety, and discrimination. This study offers a multidimensional analysis of X (formerly Twitter) discourse on Venezuelan migration in Colombia between 2014 and 2022, using natural language processing, geolocation inference, sentiment analysis, and manual annotation of derogatory terms. Drawing from 7.4 million posts, we examine how public opinion evolves across time, space, and language. While negative sentiment is pronounced in original tweets during periods of political and social unrest, our findings highlight that users tend to amplify neutral and positive messages—over 67% of retweets fall into these categories—suggesting an informal counterbalance to polarized discourse. Additionally, the analysis of the derogatory terms *veneco* and *veneca* reveals significant gendered patterns: veneca is disproportionately associated with sexualized and moralized narratives, indicating the intersection of xenophobia and sexism. These results underscore the dual function of social media as a site of both stigmatization and resistance, offering insights into the emotional and cultural dimensions of migration discourse in Colombia's digital public sphere.

1. Introduction

The outflow of refugees and migrants from Venezuela is one of the largest displacement crises in the world, with almost 7.89 million migrants and refugees as of November 2024, from which 6.71 million (85%) are currently in Latin America and the Caribbean, and 2.81 million (36%) in Colombia, making it the country that received the highest inflow [1]. Despite having a shared language, religion, and broad cultural heritage, this mass migration across Latin America has coincided with a decrease in migrant sentiment [2] and the formation of xenophobic, sexist, and discriminatory stances of the locals against the Venezuelan migrants [3]. These stances are influenced by ambivalent and even contradictory opinions and information from social networks, triggering xenophobic behaviors and actions [3]. These mixed attitudes underscore the importance of understanding how Venezuelan migration is being discussed and constructed in the public sphere.

Social media, particularly X (formerly known as Twitter), has emerged as a key arena where these attitudes and narratives play out. Users in X often engage by posting updates (posts, formerly known as ‘tweets’) and sharing others’ updates (reposts, formerly ‘retweets’). Given X’s prominence in Colombia’s media landscape, the platform offers a rich data source for examining real-time social discourse around migration. Posts can amplify both supportive voices and hostile sentiments, shaping public opinion and even policy responses. Prior research on immigration discourse indicates that social media often serves as an “echo chamber” for polarized views, where migrants may be alternately portrayed with empathy or subjected to derogatory stereotypes [4, 5]. There is a need for nuanced, multidimensional analysis that captures not only the content of posts but also their spatial [6] and cultural dimensions – for example, how conversations differ across regions within Colombia, how xenophobic language and ethnic slurs emerge in digital discussions, and whether gendered patterns or sexist tropes inflect the discourse.

This article addresses that gap by conducting a multidimensional analysis of X’s discourse on Venezuelan migration in Colombia. We combine natural language processing (NLP) techniques with spatiotemporal analysis to explore the complex ways migration is discussed online. Our study seeks to analyze the discourse surrounding Venezuelan migration by examining it across three key dimensions: time, space [6], and the use of xenophobic terms *veneco* and *veneca* [7]. In particular, we examine the thematic content and sentiment of the discussions, and the geographic distribution of these conversations. By integrating these dimensions, we aim to shed light on how public opinion is being conveyed on social media and how to identify the dominant narratives –solidarity, hostility, or ambivalence –formed around Venezuelan migration. The findings not only contribute to the scholarly literature on migration and digital discourse but can also inform policymakers and civil society on the dynamics of online sentiment in Colombia’s migration context. In what follows, we first provide background on the Venezuelan migration crisis and outline relevant literature on social media discourse, NLP methods, and the use of derogatory language before presenting our methodology and results.

2. Background and Literature

2.1. Venezuelan Migration

The collapse of Venezuela’s economy and political stability over the past decade has triggered an unprecedented migration crisis in Latin America. Since around 2015, millions of Venezuelans have fled severe shortages of food, medicine, and basic security at home (see Figure 1). As noted above, this diaspora reached approximately 7.2

million people by 2023, constituting one of the largest displacement events in the world [8]. Colombia – sharing a lengthy border with Venezuela – has borne the brunt of this exodus. By the end of 2024, roughly 2.81 million Venezuelan migrants and refugees moved to Colombia [1] This influx has raised significant humanitarian and socio-economic challenges for the host country, including strains on public services, the informal labor market, and local communities.

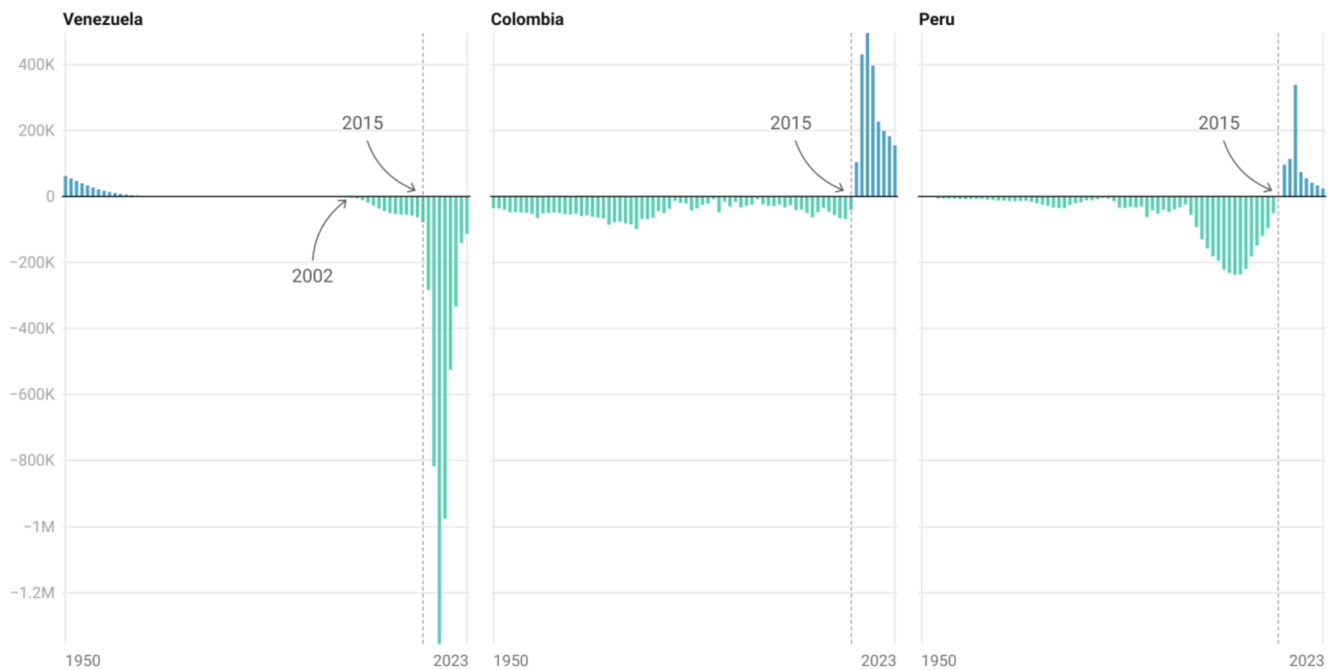

Figure 1. Net number of migrants (The total number of immigrants minus the number of emigrants) over the past 70 years. Data source: UN, World Population Prospects (2024)[9].

Colombia's official response to the crisis has been widely regarded as generous and pragmatic. Successive administrations put in place special permits and, notably in 2021, a ten-year Temporary Protected Status to regularize Venezuelans' stay and facilitate their access to healthcare, education, and formal employment (in Spanish: Estatuto Temporal de Protección para Migrantes Venezolanos or ETPV). These policies reflect a regional solidarity narrative rooted in historic ties and a sense of shared responsibility that Colombia's leaders have often invoked. For example, former President Iván Duque publicly urged Colombians to reject xenophobia and instead respond to the migrant influx with compassion and solidarity, framing it as Colombia "knowing how to confront migration crises with a sense of brotherhood". Early indications from President Gustavo Petro's government, the current president of Colombia as of 2025, suggest a continuation of this welcoming stance. Such political and institutional support has distinguished Colombia's approach in comparison to some other Latin American countries.

Nevertheless, the social reception of Venezuelan migrants in Colombia is complex and, at times, contradictory. Surveys and reports have highlighted growing wariness and negative stereotypes among segments of the Colombian public, especially in areas most affected by the sudden population increase [8, 10, 11]. This is particularly important, considering that Venezuelan migrants in Colombia are unevenly distributed, with heavy concentrations in major urban centers. As of mid-2022, about 21% of Venezuelan migrants were in Bogotá and 14% in Antioquia, followed by significant shares in border/state capitals like Norte de Santander (~9%, Cúcuta) and Atlántico (~7%, Barranquilla) [12]. Every department in Colombia now hosts some Venezuelan population, though migrants predominantly settle in large cities and economic hubs.

Fears about job competition, crime, and cultural differences have fueled resentment in specific communities. As Bulley [8] observes, despite the state's hospitable policies, many Colombians have expressed rising xenophobic sentiments and distrust toward the newcomers, perceiving them as a burden or security threat. Incidents of discrimination have been reported in daily life – from housing and employment discrimination to verbal harassment. In sum, while Colombia's government has embraced an open-door policy, migrants' lived experiences vary: acceptance and goodwill coexist with episodes of xenophobia and social tension. This duality makes the public discourse around Venezuelan migration a critical area of study, as it may influence integration outcomes and the sustainability of Colombia's humanitarian approach.

In this context, the *Barómetro de Xenofobia* (Barometer of Xenophobia) [10] emerged as a valuable initiative for systematically tracking how these tensions manifest in the digital sphere. Led by Fundación Otraparte and Fundación Interpreta, the project applied Big Data analytics to monitor online discourse—especially on X and Facebook—about Venezuelan migrants across Colombia's major cities. By examining how public conversations evolve around key themes like employment, public services, and security, the Barómetro identified recurring patterns of scapegoating and harmful stereotypes with narratives often portraying migrants as security threats or economic burdens, particularly during moments of political or social stress. Notably, the project conducted regional comparisons of xenophobic discourse, identifying significant variations in hostility levels across Colombian cities and departments, which helped map the geographical contours of anti-migrant sentiment. Its findings underscore the importance of digital narratives in shaping public perception and reinforce the need for evidence-based policy responses to address misinformation and mitigate xenophobic attitudes in both online and offline environments.

2.2. Use of X and Natural Language Processing

In the digital age, social media platforms serve as barometers of public opinion and arenas for contesting narratives about migration. X is especially pertinent given its popularity in Latin America and its use by politicians, news outlets, and citizens alike to discuss current events [13]. The platform's open nature and real-time spread of information mean that it both reflects and shapes public discourse. Therefore, Analyzing X content can provide insights into prevailing attitudes toward Venezuelan migrants and how those attitudes fluctuate or respond to events. Indeed, a growing body of research has employed computational text analysis and NLP techniques to study immigration-related conversations online. For example, Ekman [4] documented how anti-immigration actors leverage social media to propagate racist and exclusionary narratives, finding that user interactions often construct immigrants as cultural and security threats in Europe's digital sphere. Similarly, studies in other contexts have found that X posts about refugees frequently polarize around opposing values – humanitarianism versus nativism – especially following high-profile policy changes or incidents [5].

These findings suggest that the conversation in X in Colombia might also host competing framings of Venezuelan migration, from empathetic messages welcoming refugees to hostile rhetoric blaming migrants for social ills.

Our study harnesses NLP methods to examine such content at scale systematically. By collecting a large corpus of posts mentioning Venezuelan migrants (e.g., using keywords like *venezolanos* or specific hashtags), we applied algorithms to identify dominant topics and sentiments. Sentiment analysis, for instance, helps quantify the emotional valence of posts (positive, negative, or neutral) and track how sentiments change over time or in reaction to events [14]. Topic modeling or keyword frequency analysis can reveal common discussion themes – whether posts focus on humanitarian concerns, security issues, economic impacts, or political debates. Importantly, we incorporate a geospatial lens in our analysis. X data can be enriched with location information either through geotagged coordinates or user-provided location in profiles. This allows us to approximate where posts are coming from and thus examine geographic variations in discourse. For example, do border regions (which experience the most direct impact of migration) show a higher intensity of certain narratives than interior regions or the capital? Previous research cautions that only a small subset of posts are explicitly geotagged, and the user-declared locations can be noisy or ambiguous [15]. Nonetheless, scholars have developed methods to infer user locations and have successfully used geo-referenced X data to study human mobility and local attitudes. Armstrong et al. [15] note both the promise and pitfalls of using geolocated posts in migration research: while such data can reveal migration flows and community reactions in near real-time, biases exist since younger, urban, and digitally savvy populations are overrepresented on X. We mitigate these issues by applying data cleaning steps and cross-referencing multiple location cues from the posts.

Using NLP and location-based techniques, our analysis aims to capture a multidimensional view of the discourse. This entails not just what is being said (content and sentiment), but also where it is being said (spatial context). Such an approach follows an emerging trend in migration studies that leverages big data and computational tools to complement traditional surveys and media analyses. It enables the processing of thousands of posts to detect patterns that manual qualitative analysis might miss. However, we remain attentive to context – for instance, understanding Colombian cultural references or slang in posts is crucial for accurate interpretation. In the next section on derogatory terms, we delve deeper into specific language features of the X discourse that carry social significance.

2.3. Derogatory terms

A central concern in analyzing discourse around migrants is the presence of derogatory language, including ethnic slurs and other forms of hate speech. Such language is not merely offensive; it serves as a barometer of xenophobia in society and can actively contribute to the marginalization of the targeted group [16]. In the case of Venezuelan migration, a number of pejorative nicknames and stereotypes have circulated in host communities, and these inevitably find their way into online discussions. For example, Venezuelan migrants are sometimes referred to as *venecos* – a derogatory Spanish slang term – or other insulting labels that underscore their outsider status. The use of these slurs on X provides a stark indicator of anti-immigrant sentiment. Identifying and analyzing these terms in our dataset allows us to measure the extent of hostile discourse and to investigate the contexts in which xenophobic expressions spike (such as in response to news events or misinformation). Prior studies have underscored that derogatory framing of migrants on social media often goes hand in hand with false or exaggerated narratives of criminality and economic threat, reinforcing a climate of fear and intolerance. For instance, Aldamen [5] found that on Turkish and Jordanian social media, the negative portrayal of Syrian refugees – through otherizing labels and unverified rumors – significantly fueled hatred towards them, leading to psychological distress among refugee communities. This suggests that the rhetoric used to discuss migrants has tangible effects: it can shape public perceptions, justify discriminatory practices, and even influence policy if hateful narratives gain enough traction.

In the Colombian context, the intersection of xenophobia with sexism is an additional dimension that warrants attention. Social media discourse can sometimes target Venezuelan women in gender-specific ways, reflecting broader misogynistic tendencies. Reports have noted how Venezuelan women are stereotyped or demeaned in online commentary – for example, being hypersexualized or accused of “stealing” jobs and spouses – which combines xenophobic resentment with sexist tropes. Close to half of the people consulted in a study in Colombia, Ecuador, and Perú think that migrant women will end up engaging in prostitution [3]. In fact, while *veneco* slang for a Venezuelan man is used as an insult, *veneca* has outright become a synonym for prostitute *[17]*.

This mirrors patterns observed in other parts of the world. Vochocová [18] describes how online discussions about immigration in Europe often become arenas for intersectional hate, where racism and sexism intermingle. In her analysis of Czech-language forums, she noted cases where female figures (whether immigrant women or local women allied with immigrants) were disparaged using gendered insults alongside anti-immigrant rhetoric, revealing a toxic nexus of prejudices. Translating this insight to our study, we are attentive to any gendered epithets or double standards applied to Venezuelan migrant women in the X discourse. For example, if female migrants are more frequently mentioned in contexts of prostitution or blamed for social problems differently than male migrants, it would indicate a gendered pattern of xenophobic discourse.

While it can seem like an exaggeration to refer to slang, like *veneco/a*, as discriminatory xenophobic rhetoric, it is appropriately classified as hate speech by meeting the following markers: (1) it provokes hatred and violence to large populations; (2) uses social media as medium; (3) is used broadly to describe groups; and (4) often used in rhetoric [19]. Scholars of linguistics, culture, and communication have started to take notice and offer an explanation for this shift. A sociologist at Universidad del Norte explains that when people are grouped with an adjective, even if it is a word derived from their nationality, it serves as a conduit of stigmatization and rejection. Now that there is a negative image about Venezuela, referring to Venezuelans even with a word like *veneco* derived as a shortened term for their nationality, will reproduce a negative association [20]. In this way, the word changes from a colloquialism without a predetermined connotation to a pejorative.

However, the use of these terms has not always been the same; they have changed because of major political and economic events in the late 20th century. During the 1970s, Venezuela was a safe haven for Colombians fleeing widespread civil conflict in hopes of work opportunities and a better life [21]. This amicable relationship and relatively porous border allowed Colombians to traverse between border towns with little immigration enforcement. During the 1970s when Venezuelans hosted millions of Colombians, the words *veneco* (masculine) and *veneca* (feminine) "weren't always taken as an offense by Venezuelans." Sergio Chacón, Master in Linguistics and Spanish, explains that this word arose from a kind of cultural and linguistic syncretism born within the border between both countries" [20]. The term, in its inception described Colombians that lived in Venezuela and developed a new accent, whereas now, it is a descriptor for Venezuelans migrating to Colombia [3]. This makes it interesting to understand how these terms evolved over the years.

Therefore, by examining the frequency and context of derogatory terms in our X corpus, we contribute to understanding the prevalence of hate speech in this migration debate. We also analyze co-occurrence patterns – i.e., what other words or topics tend to appear in posts that use slurs. This helps illuminate whether such posts are isolated outbursts of name-calling or part of more coherent narratives (for instance, framing migrants as criminals or invoking nationalist sentiments). The literature on hate speech in digital spaces emphasizes that these expressions are not random; they often align with broader nativist discourses and can be exacerbated during times of social stress or political rhetoric that scapegoats migrants. Moreover, we consider the counter-discourse: alongside derogatory posts, are there users actively pushing back against hate speech, calling out xenophobia, or using humor/solidarity hashtags to humanize Venezuelans? The presence of such counter-narratives would indicate an ongoing contestation in the online space, which is itself an important aspect of the discourse dynamics.

Existing research on migration discourse and social media provides valuable frameworks – from understanding the role of values and emotions in posts to recognizing how hate speech proliferates in networked publics [4, 5] to appreciating the layered prejudices that can surface (xenophobia entwined with sexism or other biases) [18]. Building on these insights, our study applies a multi-faceted analytical lens to a large X dataset, aiming to map the discursive landscape surrounding Venezuelan migrants. The following sections will detail our data collection and methodology before presenting the results of our analysis on the content, sentiment, spatial distribution, and linguistic characteristics (including derogatory terms) of the X discourse. We will then discuss what these findings reveal about the broader social discourse in Colombia and the implications for migrant integration and social cohesion.

3. Methodology

The proposed methodological framework includes four sequential stages, as outlined in

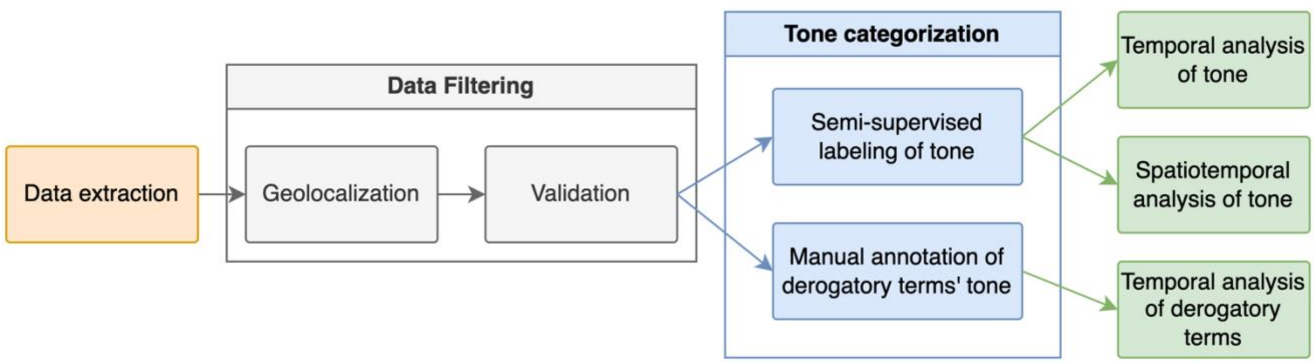

Figure 2. These stages include (1) the extraction of posts; (2) filtering by geolocation and thematic relevance to Venezuelan migration; (3) the categorization of post tone through a semi-supervised model and manual annotation of derogatory terms' tone; and (4) a multi-level analysis consisting of temporal, spatiotemporal analyzes of the tone of the conversation, and a temporal analysis of the derogatory terms *veneco* and *veneca.*

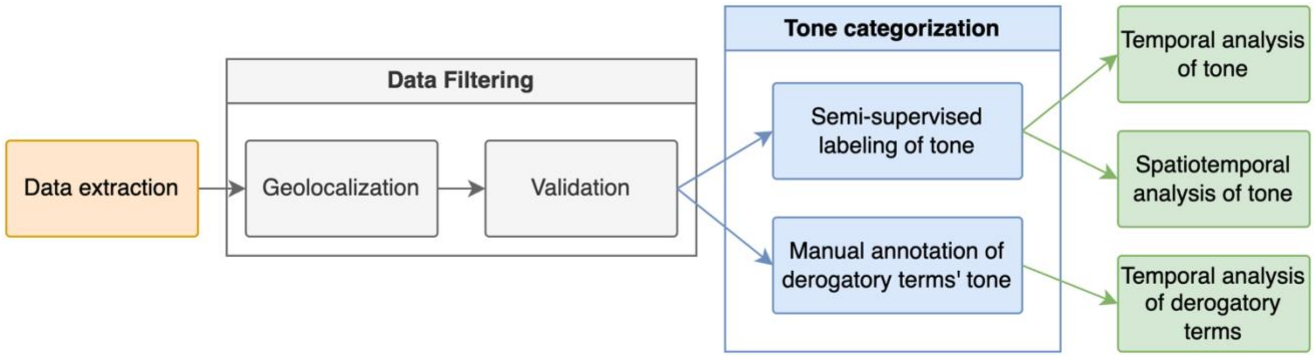

Figure 2. Methodology.

3.1. Data extraction

Data was collected using the former Twitter Streaming API, we ran a search query containing keywords related to the Venezuelan migration. The query was designed to retrieve content that refers to *migration* or *migrants*, in conjunction with terms identifying Venezuelan nationality (including *venezolano/a/os/as*, *Venezuela*, and common abbreviations such as *VZLA* or *VNZLA*), or that contains colloquial and often derogatory labels such as *veneco*, *veneca*, *venecos*, or *venecas*, accounting for plurality and grammatical gender (feminine and masculine) in Spanish. The search retrieved not only original posts but also reposts and mentions referencing the target keywords. This approach yielded a dataset comprising 7.4 million posts, spanning the period from January 2014 to June 2022.

3.2. Data Filtering

3.2.1. Geolocalization

To ensure that the dataset was composed exclusively of posts relevant to Venezuelan migration, we developed a binary classification model capable of distinguishing between pertinent and non-pertinent content. A manually labeled subset of 4,684 posts was created, wherein each post was reviewed independently by human annotators and assigned a binary label: relevant (i.e., discussing Venezuelan migration) or irrelevant (i.e., unrelated to the topic). This initial annotation effort resulted in 2,990 posts classified as relevant and 1,693 as irrelevant.

From the extracted dataset, posts were first filtered by geographic origin. Of the 7.4 million posts, only 2.35 million had active geolocation data as user-defined location fields, while the remaining 5.1 million had no data associated. The user-defined locations are free-text text with no discrimination of geographic levels, such as countries or cities. For instance, in “Bogotá,” it is not explicitly mentioned that it is within the country of Colombia. We employed the Nominatim API [22] to standardize user-provided locations to identifiable geographic entities (countries/nations).

Of the 2.35 million, the most frequent countries were Venezuela (47%), Colombia (28.4%, with 353,776 posts), the United States (16.9%), and Spain (14.2%). As we wanted to capture the whole conversation in Colombia, we trained a classification model to determine whether a post was from Colombia based on its text, and used it to automatically label the remaining 5.1 million with no user-defined location

For this, we fine-tuned two Spanish-language NLP models—spaCy's es_core_news_sm [23] and BETO [24], a BERT variant trained on Spanish corpora—using a training dataset of 270,000 posts (135,000 from Colombia and 135,000 from other countries), a sample taken from the posts geolocated with the Nominatim API. For all the models in this project we created the training and testing datasets using a 70/30 split, this ratio is commonly used for classification tasks with neural networks, [25] showing that it provides the best performance compared to other ratios in their task. The spaCy model achieved the highest classification accuracy at 0.79 and was therefore selected to infer geolocation for the remaining posts. The training results are presented in Table 1. Ultimately, 2 million posts were identified as originating from Colombia.

Table 1. Performance of the text geolocation models.

Architecture	Accuracy
es_core_news_sm	**0.79**
BETO	0.50

3.2.2. Validation

To ensure that our analysis focused exclusively on posts relevant to the Venezuelan migration, we developed a classification model to distinguish relevant from irrelevant posts. We began by extracting a subset of 4,684 posts from our dataset for manual labeling, then training the model and ultimately using it to automatically label the 1.65 million posts from Colombia. Each post was independently reviewed and categorized as either relevant (discussing Venezuelan migration) or irrelevant (unrelated topics). The labeling resulted in 2,990 posts marked as relevant and 1,693 as irrelevant.

To prepare the data for model training, we addressed class imbalance and duplication issues. We removed duplicate posts from the relevant category and performed random equal balancing to create a balanced dataset. This process yielded 1,554 posts in each category, totaling 3,108 posts. The dataset was then partitioned using a 70/30 train-test split, with 2,170 posts allocated for training and 938 for evaluation.

Three models were fine-tuned for this task: spaCy's es_core_news_sm, the BETO transformer model, and bert-base-multilingual-uncased-sentiment [26]. Performance was assessed using classification accuracy on the held-out test set and are presented in Table 2. Among the models evaluated, spaCy's es_core_news_sm exhibited the highest accuracy (0.90), outperforming both BETO (0.81) and the multilingual BERT variant (0.84). Due to its superior performance, spaCy's model was selected to automatically label the remaining dataset, ensuring a consistent and high-quality identification of migration-relevant posts across the corpus. Ultimately, 1.65 million posts were identified as relevant.

Table 2. Performance of the Validation models.

Architecture	Accuracy
es_core_news_sm	**0.90**
BETO	0.81
BERT-base multilingual-uncased-sentiment	0.84

3.3.Tone and topic categorization

3.3.1. Semi-supervised labeling of tone

We employed a semi-supervised learning approach built upon a manual annotation to evaluate the sentiment expressed in posts about Venezuelan migration. A subset of 2,990 posts previously classified as relevant was annotated for tone by two independent coders. Each post was assigned a score on a 7-point Likert scale ranging from -3 (extremely negative) to +3 (extremely positive), with 0 representing a neutral tone. The labeling rubric was refined through iterative discussion among the research team to ensure conceptual clarity and coding consistency (See Figure 3).

Negative			Neutral	Positive		
-3	-2	-1	0	1	2	3
destructive and/or xenophobic	harmful and/or pessimistic	opposing	unbiased	in favor	benefical and/or affirmative	constructive and/or anti-xenophobic
Opinions agains the subject or adjectives to describe them; expressed with xenophobic, insulting, or derogative words.			No opinion expressed	Opinions in favor of the subject or adjectives to describe them; expressed with open-minded, benefical, or supportive words.		

Figure 3. Criteria used for the tone scale labeling.

Given the relatively limited size of the labeled dataset, sentiment scores were ultimately grouped into three categories: negative (comprising -3 to -1), neutral (0), and positive (1 to 3). This aggregation enabled the creation of a more robust classification task, minimizing sparsity and class imbalance. The intercoder agreement was calculated at 63.5%, which, while moderate, aligns with standards in computational social science where sentiment coding is subject to varying interpretations and contextual nuances [27]. For all posts in which coders initially disagreed, a consensus labeling was reached through in-person reconciliation sessions, during which coders collaboratively resolved discrepancies and clarified ambiguous cases.

The final labeled corpus contained 1,920 posts, which was split into training and testing subsets using a 70/30 ratio. We trained and evaluated three classification architectures on this task: es_core_news_sm, bert-base-spanish-wwm-cased, and bert-base-multilingual-uncased-sentiment. As shown in Table 3, the Spanish BERT model achieved the highest accuracy (0.699) and was subsequently used for labeling the full set of migration-related posts. This model was selected for its superior performance and its pretraining on large-scale Spanish corpora, which allowed for more accurate interpretation of nuanced sentiment in the dataset.

Table 3. Performance of the tone classification models.

Architecture	Accuracy
es_core_news_sm	0.552
bert-base-spanish-wwm-cased	**0.699**
bert-base-multilingual-uncased-sentiment	0.666

3.3.2. Manual annotation of derogatory terms' tone

In parallel with the general tone classification, we conducted a targeted manual annotation of posts containing the terms *veneco* and *veneca*, which have increasingly been associated with pejorative or xenophobic discourse in the Colombian context. This analysis aimed to capture these terms' evolving connotations and affective load in digital conversations. Two independent coders reviewed and labeled each post containing either term, applying the same 7-point Likert scale used in the broader sentiment analysis (see Figure 3). The criteria were adapted to focus on how the terms were used in reference to Venezuelan migrants, whether as neutral descriptors or as explicit slurs.

A total of 1,175 posts were annotated for this task, with 870 containing veneco and 305 containing veneca. Following the same aggregation procedure as the general tone classification, the scale was collapsed into three categories: negative, neutral, and positive. Due to the limited number of posts involving these terms, this grouping allowed for greater analytical coherence while still preserving meaningful distinctions in sentiment. Intercoder reliability for this task was calculated separately for each term: 59.7% for *veneco* and 61.4% for *veneca*. While modest, these levels of agreement are consistent with existing research on manual sentiment labeling, particularly for emotionally charged or context-dependent expressions. The annotated data was subsequently used in descriptive temporal analyses of pejorative language, as presented in later sections of the paper.

4. Results

4.1. Temporal analysis of tone

The analysis of post tone from 2014 to 2022 reveals notable differences in sentiment distribution when considering original posts alone versus including reposts (Figure 4 and Figure 5, respectively). Without reposts, the tone profile of the discourse skews more negative; however, the inclusion of reposted content markedly increases the proportion of neutral and positive tone posts. In fact, approximately 67% of all reposts in the dataset carry a neutral or positive tone, meaning only about one-third are negative. This indicates that users disproportionately share (repost) content that is not negative, effectively diluting the prominence of negative sentiment in the overall conversation.

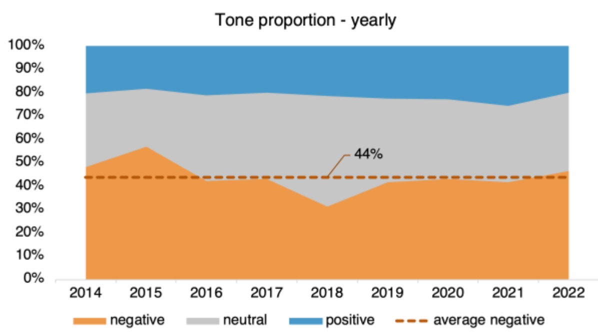

Figure 4. Tone yearly proportion.

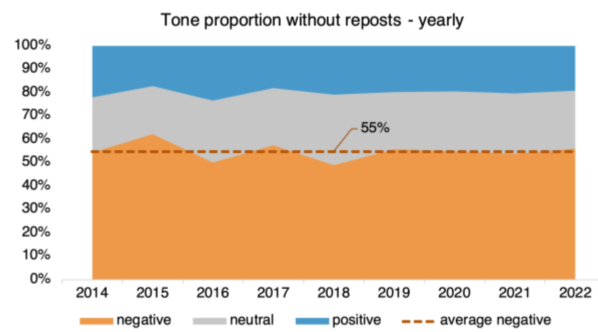

Figure 5. Tone yearly without reposts.

Our findings suggest that reposting often serves as a form of endorsement or agreement, thereby amplifying messages deemed agreeable or informative. Content with a calmer or more constructive tone tends to gain more visibility through reposts. As a result, the aggregate sentiment on X appears more neutral-to-positive once repost behavior is accounted for. This stabilizing effect of reposts implies that the public discourse on social media is co-shaped by audience amplification choices, not just by the tone of original posts. In our context, while contentious issues generated plenty of negative posts, the community's sharing behavior favored neutral or uplifting content, helping to moderate extreme sentiment in the visible discourse.

The tone of the discourse varied considerably over time, with key inflection points aligning closely to sociopolitical developments. From 2014 to 2016, post volumes were relatively low, and sentiment was more evenly distributed across tones. Neutral posts, often driven by news reporting or informational content, formed the plurality, while negative posts were less prevalent. As shown in Figure 6 and Figure 7, the years 2017 and 2018 marked a shift: negative posts became more prominent, particularly leading into the 2018 Colombian presidential election. During this period, migration emerged as a major political issue, often framed in alarmist terms [28]. The discourse grew increasingly polarized, with rising negativity evident in both raw post counts and proportions. However, much of the content that gained visibility through reposts retained a more moderate tone, highlighting the stabilizing role of amplification behaviors observed in Figure 4.

In the years that followed, specific events produced sharp changes in sentiment. A notable spike in negativity occurred in November 2019, coinciding with the expulsion of 59 Venezuelans accused of inciting unrest during mass protests [29]. This event, widely discussed online, contributed to a surge in hostile sentiment. The onset of COVID-19 in early 2020 introduced additional volatility, as pandemic-related anxiety and misinformation fueled further negativity. In contrast, February 2021 saw a pronounced increase in positive sentiment following the announcement of the Estatuto Temporal de Protección para Migrantes Venezolanos (ETPV), a policy widely celebrated in domestic and international circles. While negative tones reappeared in late 2021 and 2022, the overall sentiment began to stabilize. Reposts continued to play a moderating role throughout, reinforcing a more balanced representation of public discourse even during periods of heightened emotional intensity.

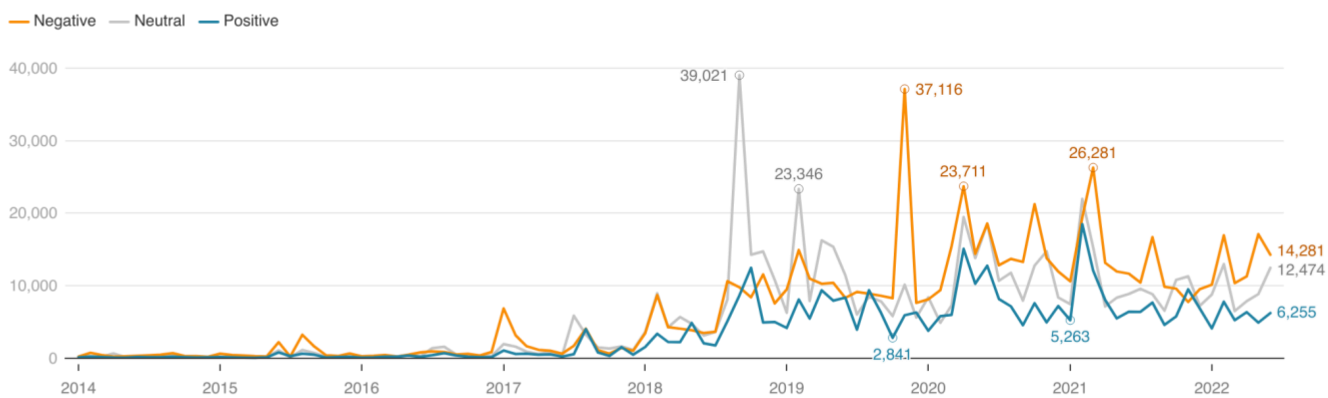

Figure 6. Tone monthly frequency.

When analyzing only original posts—excluding reposts—reveals a consistently higher proportion of negative sentiment across the entire study period (see Figure 7). This dataset version captures more unfiltered public opinion, reflecting what users choose to post rather than amplify. Notably, the share of negative posts is substantially higher in this view, particularly during politically and socially charged moments such as the 2019 expulsions and the peak of the COVID-19 pandemic mid-2020. Unlike the broader dataset that includes reposts, the original posts display more abrupt and extreme shifts in tone, suggesting that emotional, and at times reactionary, expressions are more common in initial posts. This pattern supports existing findings that original content tends to carry stronger affective language, especially during crises or contentious events. The divergence between Figures 7 and 8 thus highlights a critical dynamic in online discourse: while original posts may reflect raw public sentiment, the selective amplification of more neutral or constructive content through reposts functions as a filter, shaping the overall tone of public conversations that reach broader audiences.

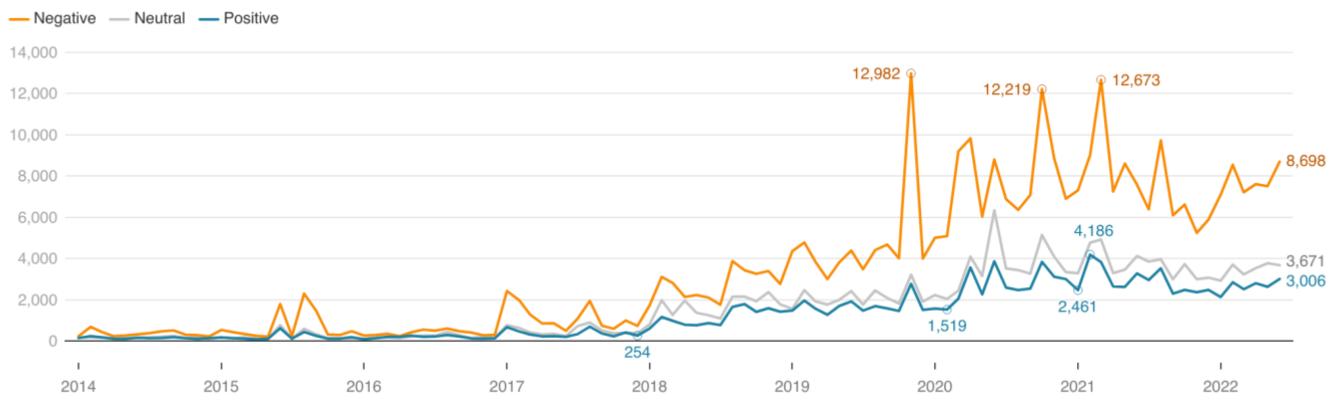

Figure 7. Tone monthly frequency without reposts.

These temporal patterns, especially the divergence between original posts and reposted content, offer valuable insight into how digital discourse is constructed and maintained. While original posts provide a direct view into users' immediate reactions—often more emotional and polarized—reposts serve as a form of social curation, selectively amplifying content perceived as more legitimate, informative, or acceptable within broader public norms.

Although negative messages tend to spread more quickly, positive messages often reach larger audiences—a pattern suggesting that users are more inclined to share or engage with uplifting or affirming content. Moreover, sentiment dynamics vary across different types of events: positive sentiment tends to accumulate gradually around highly anticipated developments, while unexpected or disruptive events typically trigger immediate surges in negative sentiment [30]. This pattern aligns with our findings, where moments of heightened negativity—such as the 2019 expulsions or the early phases of the COVID-19 pandemic—were accompanied by a predominance of more neutral or constructive content in the reposted layer of discourse. These dynamics demonstrate that affective polarization in original posts can coexist with counterbalancing mechanisms in public discourse, particularly through users' selective amplification of moderated content and the prominent role of institutional actors whose posts are frequently reposted.

At the same time, the influence of real-world events remains predominant in shaping digital sentiment. Our results align with prior research establishing X as a barometer of public mood, sensitive to crises, political milestones, and policy shifts. For example, the clear spike in positive sentiment during the announcement of the ETPV policy in February 2021 underscores how online tone can shift rapidly in response to policy changes perceived as equitable or humanitarian. Conversely, moments of heightened insecurity or political unrest, such as the November 2019 deportations, leave visible imprints in the form of sudden increases in negative sentiment.

The literature on social stress theory and scapegoating provides insights into how societal instability and economic fear can lead to the externalization of public anxiety and attribution of personal or social problems toward vulnerable groups, particularly migrants [31, 32]. Our data capture this phenomenon, as sentiment fluctuates not in a vacuum but in reaction to unfolding political and social contexts. This reinforces the idea that public discourse online is both a reflection and amplification of the national mood, mediated by the immediacy and virality of platforms like X.

Ultimately, this study contributes to a more nuanced understanding of how sentiment evolves in digital environments. Rather than a linear or unidirectional shift toward hostility or empathy, the data reflect a dynamic balance: spikes in sentiment driven by exogenous events are tempered by endogenous mechanisms like repost behavior and the prominence of institutional messaging. The interplay between emotion, information quality, and user engagement produces a sentiment feedback loop—one where public mood influences content, but is also shaped by how that content is surfaced and shared.

4.2. Spatiotemporal analysis of tone

Prior research has examined how public sentiment towards migrants may vary across geographic regions. For instance, Lebow et al. [33] compared anti-migrant sentiment across regions with varying exposure to Venezuelan refugees in seven Latin American countries. Using survey data and geotagged social media posts, they applied statistical comparisons—including an instrumental variable approach—at multiple geographic scales. Their analysis found no evidence that regions hosting more migrants exhibited more negative sentiment. Rather, anti-immigrant attitudes appeared to be shaped predominantly by national-level discourses and narratives, primarily disconnected from local experiences or pressures such as labor competition, service strain, or crime rates.

In a more localized study, Mora et al. [34] conducted a spatial analysis of migration in Cali, Colombia. Using spatial econometric modeling, they identified a positive spatial correlation between immigrant unemployment and crime levels at the neighborhood level. This suggests that in some urban contexts, socio-economic stress associated with migrant integration may contribute to localized security concerns.

Building on this literature, we conducted a spatiotemporal analysis of post sentiment toward Venezuelan migrants within Colombia. Our primary aim was to explore whether the tone of posts—classified as positive, negative, or neutral—varied by geographic region and whether such variation correlated with departmental-level demographic and socio-economic indicators. We determined the corresponding department using the latitude and longitude coordinates embedded in posts. This allowed for disaggregated sentiment analysis across space and time.

First, we analyzed the number of posts of each tone (positive, neutral, negative) over time and by department. For clarity in the visualizations, we focused on the six departments with the highest concentrations of Venezuelan migrants between 2018 and 2022: Bogotá, D.C., Valle del Cauca, Santander, Atlántico, Antioquia, and Norte de Santander. Figure 8 illustrates each department's monthly frequency of posts classified by tone. While all regions showed a predominance of negative sentiment, notable temporal and regional variations emerged. Bogotá D.C.'s sentiment trends largely mirrored national averages, while departments such as Valle del Cauca, Santander, and Atlántico exhibited episodic surges in negative sentiment during particular months. These fluctuations may reflect localized events, political discourse, or shifts in media narratives.

Figure 8. Tone frequency over time for the six departments with the most Venezuelan migrants.

Message amplification mechanisms may also shape regional sentiment. In line with our national-level findings, reposting behavior had a stabilizing effect on tone: including reposts reduced the relative proportion of negative posts while increasing the share of neutral ones. This suggests that messages perceived as less negative may be more widely circulated, thereby influencing the tone of the broader discourse.

To assess whether other regional variables help explain sentiment tone, we examined correlations between the proportion of posts of each tone (normalized by department population) and a set of demographic and socio-economic indicators. As an initial step, we observed strong correlations (~0.8) between the raw counts of positive, neutral, and negative posts and departmental population size. This implies that post volume is primarily driven by the number of residents, with more populous departments naturally producing more content across all tone categories. Figure 9 illustrates this relationship.

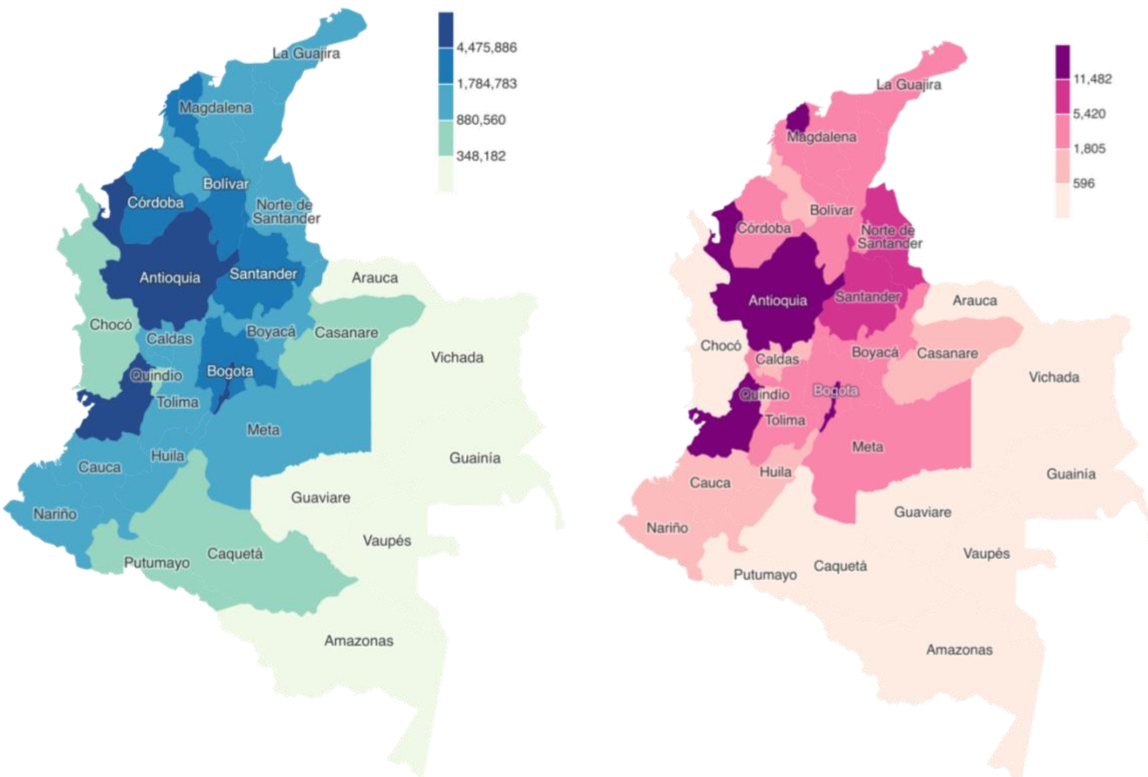

Figure 9 (a) Population of Colombia by departments in 2018 based on DANE [35], and (b) total number of posts per department.

To address this confounding effect, we normalized post counts by department population. We then conducted bivariate correlation analyses between these normalized sentiment proportions and a set of variables, including mugging incidents, unemployment, poverty, inequality, and homicide rates.

Results from these analyses are presented in Table 4. While most correlations were low in magnitude ($r < 0.46$), a few patterns deserve attention. Notably, the Gini coefficient and unemployment rate negatively correlated with the proportion of positive posts (r = -0.454 and r = -0.404, respectively), reaching statistical significance at the 5% level. These findings suggest that departments with greater income inequality and higher unemployment tend to produce fewer positive posts about Venezuelan migrants, even after adjusting for population size. While these relationships do not imply causality, they align with the broader literature suggesting that socio-economic strain may correspond with diminished receptiveness toward migrant populations.

Table 4. Correlations between variables and the proportion of posts of each tone (by department); r being correlation and p its p-value.

Variable	Positive		Neutral		Negative	
	r	p	r	p	r	p
Number of mugging incidents per department in 2018	0.287	*0.105*	-0.368	*0.035*	-0.031	*0.864*
Unemployment rate per department in 2018	0.261	*0.218*	-0.099	*0.644*	**-0.404**	*0.050*
Poverty rate per department in 2018	-0.022	*0.918*	0.153	*0.475*	-0.310	*0.140*
Extreme poverty rate per department in 2018	-0.110	*0.609*	0.103	*0.632*	0.024	*0.911*
Gini coefficient per department in 2018	0.095	*0.659*	0.093	*0.665*	**-0.454**	*0.026*
Number of homicides per department in 2018	0.261	*0.149*	-0.193	*0.290*	-0.133	*0.467*

In general, while population size strongly influences overall post volume, normalizing sentiment tone by department population enables more meaningful regional comparisons. Although low, the observed associations between inequality, unemployment, and positive sentiment suggest that structural socio-economic conditions may play a role in shaping the tone of public discourse on migration.

4.3. Temporal Analysis of Pejorative Terms

The temporal evolution of specific terms used to describe Venezuelan migrants in Colombia offers insight into how language adapts within public discourse and may contribute to processes of stigmatization. To that end, we analyzed the monthly and yearly distribution of posts containing the terms *veneco* and *veneca*, as illustrated in Figure 10.

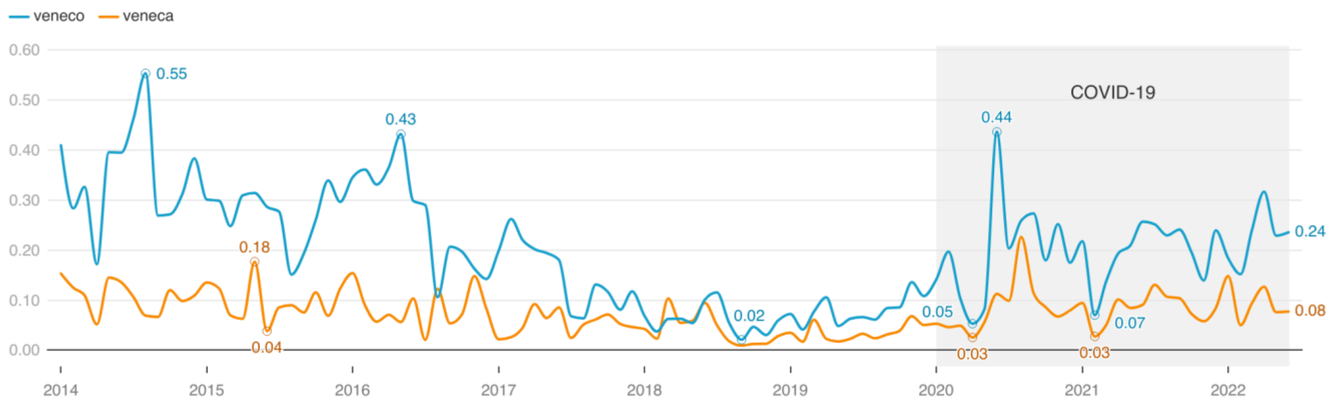

Figure 10. Monthly proportion of posts containing the terms veneco and veneca.

Throughout most periods under review, the usage frequency of veneco exceeded that of veneca, a pattern potentially attributable to grammatical conventions in Spanish—where masculine plurals encompass mixed-gender groups. This makes veneco a more general reference and, therefore, more prevalent. From 2014 through mid-2018, the proportion of posts containing veneco and veneca exhibited a generally downward trend. This is particularly notable given that during this period, Venezuelan migration into Colombia had not yet reached its peak. Before 2015, the use of these terms was not exclusive to Colombian speakers referring to Venezuelans; rather, they were sometimes used reciprocally, with veneco and veneca serving as regional colloquialisms without uniformly pejorative connotations [20]. The steady decline in usage between 2015 and 2018 suggests a moment of semantic ambiguity, in which the terms had not yet fully transitioned into overtly xenophobic labels in Colombian discourse.

The period following mid-2018 marks a transition in both migration patterns and political context. Colombia experienced a sharp rise in the number of Venezuelan arrivals—surpassing one million that year— (see Figure 1) as economic and political instability in Venezuela intensified [36]. This migration wave coincided with the Colombian presidential elections, in which migration policy became a salient campaign issue, and a political and xenophobic narrative was constructed and positioned by the candidates and the media [28]. Despite the substantial demographic shift and the politicization of migration, the frequency of veneco and veneca posts remained relatively stable between 2018 and 2020. This may reflect a plateau in the social salience of the terms as they became embedded in the online discourse, neither intensifying nor subsiding significantly.

However, a renewed increase in the usage of both terms is observable beginning in early 2020, reaching marked peaks in mid-2020 and again in 2021. This resurgence aligns with the onset and escalation of the COVID-19 pandemic in Colombia, a period characterized by severe economic contraction, unemployment, and increased strain on public health systems. These conditions, as evidenced in prior studies [37, 38], often create fertile ground for scapegoating and intergroup hostility. The reintensification of *veneco* and *veneca* during this period may thus be interpreted as a reflection of mounting societal anxiety, with migrants positioned as symbolic targets within broader narratives of crisis and scarcity.

The cyclical nature of the terms' usage—declining during relatively stable periods and rising during moments of socio-political disruption—points to their evolving function within Colombian digital discourse. Rather than acting as consistent descriptors, *veneco*, and *veneca* appear to operate as affective markers whose prominence fluctuates in response to external pressures. This underscores the utility of temporal analysis in tracking the discursive transformation of pejorative terms and highlights how crises such as pandemics or political transitions can reinvigorate latent xenophobic language.

To assess whether the use of these terms carried a negative emotional tone, we analyzed the monthly proportion of posts containing veneco and veneca that were classified as negative in tone (Figure 11). Tone here refers to the emotional valence of the post—ranging from hostile or derogatory (-3) to neutral (0) to positive (+3)—based on manual annotation and supervised classification.

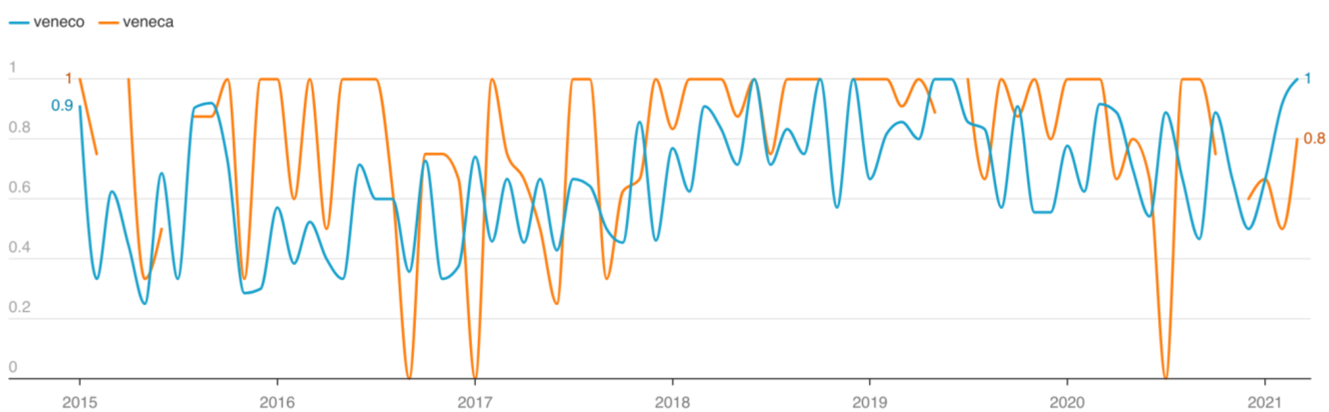

Figure 11. Monthly proportion of negative tone associated with the terms veneco and veneca.

Despite the high month-to-month variability in the tone classification—likely a consequence of smaller sample sizes, particularly for *veneca*—a broader temporal trend emerges. Between early 2015 and mid-2018, the average proportion of posts using *veneco* with a negative tone rose from approximately 54% to over 82%. Similarly, for *veneca*, the average negativity increased from around 72% to a peak of 95%. This period corresponds with the intensification of the Venezuelan migration crisis and culminates in Colombia's 2018 presidential election, during which migration was a key topic in political discourse. As discussed in the previous section, the stabilization in the use of both terms during this period—despite a substantial rise in the number of Venezuelan migrants—suggests that the tone of discourse, rather than its frequency, became increasingly hostile. The rise in negative sentiment thus reflects not just a growing presence of migrants but also a deepening polarization in public attitudes.

Following this peak, the data show a decline in the proportion of negative sentiment beginning in mid-2019. By early 2020, the average negativity associated with *veneco* had decreased to approximately 73%, while *veneca* remained higher at around 79%. Although these values are lower than their 2018–2019 peaks, they still represent a marked increase from their pre-2018 levels, suggesting a lasting semantic shift. This post-2019 period coincides with the relative normalization of migration as a political and social issue in Colombia, alongside evolving public narratives around Venezuelan integration. However, the emergence of the COVID-19 pandemic in early 2020 triggered a brief resurgence in hostile tone—particularly for *veneco*—consistent with prior literature indicating that public health crises often catalyze the scapegoating of migrants [37, 38].

Overall, the tone analysis reinforces and extends the findings from the term frequency analysis. While the absolute use of *veneco* and *veneca* fluctuated in response to contextual events, the tone of these terms became increasingly derogatory as migration intensified and political and economic pressures mounted. Notably, this shift appears more pronounced for veneca, echoing findings in gendered discourse studies that emphasize how female migrants are uniquely subject to moral and sexualized judgments [19, 39]. Thus, even as the use of these terms stabilized or declined in volume, their semantic function within digital discourse appears to have solidified as instruments of xenophobic and gendered stigmatization.

Analyzing the topics of the annotated posts revealed four general categories of negative tone: derogatory name calling, political outrage, deprivation mocking, and perceived criminal behavior or sexual deviance. While criminality was ascribed exclusively to men, sexual deviance was attributed solely to women, reflecting deeply gendered narratives in xenophobic discourse. Both terms shared the remaining themes—name calling, deprivation mocking, and political anger—across gender lines. Neutral tones primarily used the terms as ethnic descriptors, referring simply to Venezuelans in a non-evaluative way. Meanwhile, positive tones followed two main patterns: self-description by Venezuelans using the terms to express identity, and, for *veneco*, messages of welcome or unity between Colombians and Venezuelans. For *veneca*, positive posts often referred to favorable physical characteristics, such as beauty or body features.

Events and public commentary also influenced the fluctuation in tone. The term *veneco*, in particular, was negatively influenced by several high-profile episodes, including soccer matches between Colombia and Venezuela, and political incidents. In 2015, for example, heightened tensions between the governments—prompted by the deportation of Colombians from Venezuela—fueled a spike in hostile commentary. Similarly, in 2017, when

Colombia’s Vice President made public comments implying that housing would not be allocated to Venezuelans, online backlash surged, accompanied by a rise in derogatory language.

Notably, narratives that generalized all Venezuelan migrants—particularly women—as sex workers were among the most damaging and persistent. Posts suggesting that Venezuelan women were only “good for sex” or “willing to do anything” reveal not only societal discomfort with migrant vulnerability but also entrenched forms of gender-based violence and structural inequality. These narratives depict women’s migration as inherently transactional or morally suspect, reinforcing stigmatization and eroding public empathy. As documented in a related study, such stigmatization produces tangible consequences for migrant safety, well-being, and integration [40], perpetuating cycles of marginalization.

5. Conclusions

This study provided a multidimensional analysis of the X discourse surrounding Venezuelan migration in Colombia, combining natural language processing (NLP), geolocation inference, sentiment classification, and manual annotation of pejorative terms. By examining 7.4 million posts from 2014 to 2022, the research captured temporal, spatial, and linguistic dynamics that shape public perception of migrants in a digitally mediated environment.

Our findings underscore the central role that social media plays in reflecting and shaping public discourse on migration. X has functioned not only as a platform for the diffusion of xenophobic sentiment but also as a space where empathetic and neutral narratives are amplified, particularly through reposting behavior. The differential distribution of sentiment in original posts versus reposts reveals an important corrective mechanism within the digital public sphere: while original posts often express immediate, affect-laden responses—especially during crises—reposts tend to elevate more neutral or constructive content, contributing to a moderating effect on the overall discourse.

Temporally, we observed that sentiment toward Venezuelan migrants fluctuates in close alignment with real-world events, including elections, deportation incidents, and the COVID-19 pandemic. Periods of political instability and socio-economic stress were associated with sharp increases in negative sentiment, supporting broader theories of scapegoating and social stress. However, these periods were also accompanied by the amplification of neutral or positive content, indicating a discursive tension between affective polarization and normative moderation.

Spatially, our analysis revealed that the tone of discourse varies across departments, but this variation is not merely a function of migrant concentration. Instead, socioeconomic indicators—particularly inequality and unemployment—showed a modest but statistically significant negative correlation with the proportion of positive posts. This suggests that economic stressors may shape regional sentiment toward migrants, even after controlling for population size. These findings align with prior research that connects economic precarity with anti-immigrant attitudes while also demonstrating the value of integrating geospatial analysis into social media research.

The evolution and tone of the derogatory terms veneco and veneca offer further insight into the linguistic construction of xenophobia in the Colombian context. While their usage fluctuated over time, their emotional valence—particularly the prevalence of negative sentiment—intensified during moments of heightened political or economic pressure. Notably, the term veneca carried a consistently higher proportion of negative and gendered connotations, often used to sexually objectify Venezuelan women. This gendered dynamic reflects broader patterns of intersectional discrimination, where xenophobia is compounded by sexism and moral judgment.

Together, these findings illustrate that a complex interplay of structural conditions, affective responses, and platform-specific dynamics shape online discourse about Venezuelan migrants. While hostile sentiments are prevalent, they coexist with narratives of solidarity, resistance to hate speech, and institutional messaging. This highlights the importance of not treating social media as a unidimensional space of hostility but as a contested arena where competing narratives coexist and influence public perception.

For policymakers and civil society, these results offer critical insights. Understanding the discursive landscape of migration can inform strategies to counter xenophobia, promote inclusive messaging, and respond to misinformation. Moreover, monitoring social media sentiment can serve as an early warning system for public mood shifts, particularly during political or economic upheaval. Future research might build on this work by exploring the role of influencers, bots, or media accounts in shaping discourse or by extending the analysis to other platforms and countries. Ultimately, this study demonstrates the power of combining computational and qualitative methods to analyze migration discourse at scale. As migration continues to be a defining issue in the region, such approaches will be increasingly essential for understanding and shaping the narratives surrounding it.

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